

Touch, facial expression of emotions, and ERP correlates

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Introduction

- Researchers have demonstrated alpha frequencies as measured by an EEG and indicative of cognitive alertness, are affected when individuals view facial expressions of emotions in pictures (Jessen & Kotz, 2011).
- Hajcak et al (2010) used event-related potential (ERP) and found increases in the P300 neural component (attention) and a vertex negative potential in N170 (processing of faces).
- The N170 component is amplified upon seeing an image of fear, happiness, disgust, or anger (Spapé et al., 2017).
- Other components (N2/P3b) also increase attention to facial expressions of emotion (Hajcak et al., 2010).
- P3 was linked to positive affect (Gable & Harmon-Jones, 2010) including facial expressions of happiness (Spapé et al., 2017).
- Touch affects alpha frequencies (Ishigaki et al., 2016) & has neural correlates, though research is inconsistent.
- Researchers found touch modulates N100 and P200 when processing pictures (Schirmer et al., 2011).
- Spapé et al (2017) reported mixed results involving touch's influence on ERP correlates but stated touch affects emotion to some extent (the late component of the P3 wave was amplified for positive affect images).
- Additionally, viewing negative emotions on faces (e.g., anger) led to increases in attentional alertness (Spapé et al., 2017).

Purpose of Pilot Study

- Address limitations of previous studies (current study participants touched longer & viewed neutral, negative, & positive emotion expressions vs. only positive & negative expressions).

Hypotheses

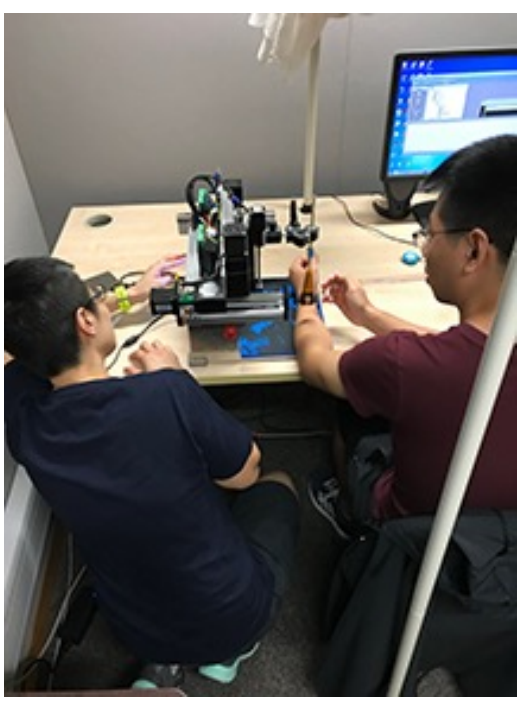
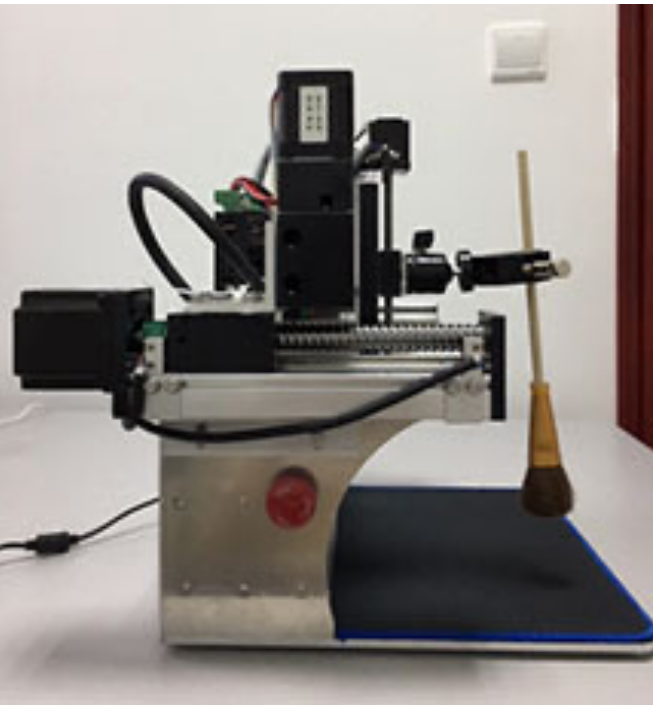
- N170 amplified when viewing fear, happiness, or anger & N2/P3b increases when viewing these emotions.
- Given previous inconsistent results, N170, N2, and P3b would not vary when participants are & are not touched.

Methods

- Data was usable from 5 U.S. Caucasian right-handed undergraduates: 3 females; 2 males.
- Participants viewed 4 unique photos of male & female actors displaying anger, happiness, fear, and neutral faces.
- The photos were drawn from a standardized dataset.
- Participants used response pad to indicate emotion they were viewing & attractiveness of person in the photo.
- Each face was randomly displayed 10 times with 32 trials per block (320 total trials/participant).
- For 160 trials, participants' forearms were lightly stroked by a brush operated by a digital touch stimulator we created, & for 160 trials they were not touched.
- The brush stroke began 10cm from the wrist & traveled 3.5cm on the forearm.
- Speed of stroke was 3.5cm per second.
- Force of stroke was .4n.
- EEG recorded with Biosemi Active Two 64-electrode cap.
- 2 external electrodes placed below the right eye to record vertical and horizontal eye movements (electrooculogram, EOG).
- 2 external electrodes placed on the left and right mastoid.
- EEG and EOG signals were acquired continuously at a 2056Hz sampling rate and referenced to the CMS reference electrode.
- All electrode impedances were kept below 40kΩ.

This study was funded, in part, by a Fulbright Scholarship awarded to Lawrence H. Gerstein. **Further information:** lgerstein@bsu.edu

Digital Touch Stimulator & Capping



Preprocessing

- EEG data were preprocessed using the EEGLAB v2021.1 toolbox in MATLAB R2017b.
- Data were resampled to 200Hz & re-referenced offline to the average of the left and right mastoid electrodes for N2 and P3b and re-referenced offline to common average for N170.
- A 0.1-50 Hz Bandpass filter was applied for line noise.
- We created an event list and extracted bin-based epochs with a time window of -200 to 1000 ms.
- Before the data onset, a baseline with a latency time of 200 ms was used for correction.
- To eliminate electromyogram & other artifacts from the ERP data, we used a rejection threshold of an extreme value of $\pm 100 \mu V$ to reject marked epochs.
- Independent component analysis was used to identify & remove stereotypical artifacts using the Runica algorithm from the EEGLAB toolbox.
- To examine N170, N2, and P3b, we analyzed the channels P7, P8, Cz, and Pz.
- Based on the inspection of the grand-average ERP waveform, we selected the time windows N170 (150-200 ms), N2 (250-310 ms), and P3b (550-650 ms).

Results

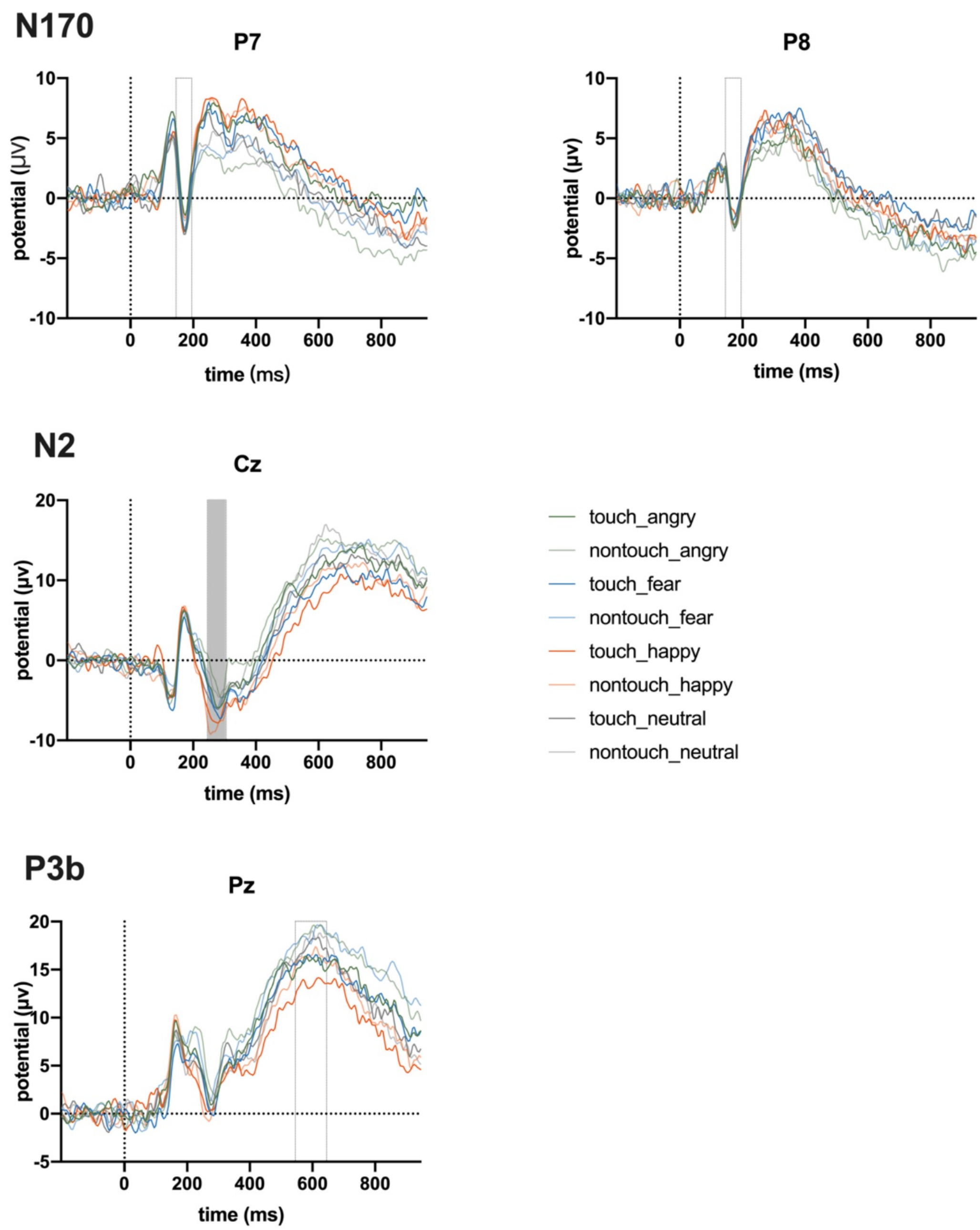
- Three 2 (touch condition: touch or no touch) x 4 (emotions: happiness, anger, fear, & neutral) within subjects MANOVAs were performed.
- One MANOVA N170 was dependent variable, another N2, and a third P3B.
- The Greenhouse-Geisser method was used to calculate the p -values for the deviations once the spherical assumption was rejected.
- If appropriate Bonferroni tests were conducted.

Details

	Reference	Time windows	Channel	ANOVA	Participants
N170	Common average	150-200 ms	P7 P8	4(Emotion)*2(touch)	5 (Excluded 25)
N2	M1&M2	250-310 ms	Cz	4(Emotion)*2(touch)	5 (Excluded 25)
P3b	M1&M2	550-650 ms	Pz	4(Emotion)*2(touch)	5 (Excluded 25)

MANOVA Results

	Main effect of Emotion	Main effect of Touch	Interactive effect
N170	ns	ns	ns
N2	$p < .001$	ns	ns
P3b	ns	ns	ns



Discussion

- No interaction found between emotion and touch for N170, N2, or P3b.
- No main effect for emotion or touch found for N170 or P3b.
- For N2: happiness (-7.5) & anger (-3.6), & happiness (-7.5) & neutral (-4.3) significantly different. More neural responses required for happiness faces compared to angry or neutral faces. No other differences for N2 and emotions.
- Small sample size a limitation.
- Perhaps intensity of emotion in faces not arousing enough.